UNI-INTEL PLATFORM OVERVIEW

Real-Time Market Intelligence & Biological Logistics Optimization for Sea Urchin Aquaculture

The **Uni-Intel Platform** is an enterprise-grade data intelligence and decision-support application custom-engineered for sea urchin farmers, specifically optimized for commercializing **Tripneustes gratilla** (Collector Urchin). Shifting the burden from manual data monitoring to automated multi-agent workflows, the platform bridges marine biology and complex international seafood logistics to transform a volatile trade into a predictable, high-margin asset.

Core Strategic Value: In the premium seafood sector, wild-harvest volatility and logistical cold-chain failure represent severe financial risks. The Uni-Intel platform continuously addresses these vulnerabilities by tracking global distribution networks, market competition gaps, and real-time transit survivability parameters to maximize farm-gate profit margins.

1. Decoupled Multi-Agent Architecture

The platform operates via three core autonomous digital agents structured inside an agentic workspace framework:

- **The Scout Agent (The Hunter):** Automates data aggregation across global trade hubs (including Japan, North America, Oceania, and Southeast Asia). It utilizes advanced Vision-OCR to extract daily wholesale pricing from scanned Japanese PDF auction sheets (e.g., Toyosu and Osaka central markets), dynamically normalizing raw currencies into standard commercial metrics: **AUD/kg** for bulk live whole urchins and **AUD/100g** for processed roe (uni) trays. It explicitly identifies target species to isolate your product from wild-caught regional competitors.
- **The Logistics Agent (The Guardian):** Actively monitors the transport viability window. The agent connects to real-time air freight schedules and weather telemetry streams at key transshipment hubs (such as Singapore, Hong Kong, and Tokyo), tracking transit duration and airport tarmac temperatures against hyper-specific biological boundaries.
- **The Intelligence Agent (The Analyst):** Acts as the primary decision engine. It ingests environmental stress data from the Logistics Agent and evaluates live batch transport safety through a peer-reviewed predictive mortality model.

2. Biological Calibration & Automated Risk Mitigation

Unlike generic logistics trackers, the platform integrates proprietary data from 3.5-year commercial-scale trials and marine aquaculture research. Transit risk is calculated continuously via a live biological regression equation:

$$R = \alpha(T - 12)^2 + \beta(t) + \gamma(\Delta W)$$

Where the optimal thermal target is anchored at ***12°C***, t is total out-of-water transit time (flagging a strict ***24-hour*** threshold), and γ represents the hardcoded fluid-loss coefficient of ***0.49***. If the Real-Time Risk Score (R) exceeds a safe ceiling of ***75.0***, the dashboard triggers an automated mitigation response:

Route Status	Risk Score (R)	Logistical Indicators	Automated Action Plan
Nominal Route	R < 75.0	Transit < 24h Hub Temp ~ 12°C–15°C	**PROCEED LIVE EXPORT** Route batch to premium international wholesale markets to secure peak margins.
High-Risk Route	R ≥ 75.0	Flight Delays > 24h Hub Tarmac Temp ≥ 25°C	🚨 **DIVERT TO LOCAL ROE PROCESSING** Halt live export instantly; divert the batch to local processing to save roe value.

3. Commercial Application Dashboard

The front-end user interface compiles these deep tracking layers into a single visual control room. It provides complete data provenance transparency, showing origin stamps (e.g., specific carrier cargo feeds or central market sheets) next to every price metric card. Farmers gain absolute clarity on international supply vacuum windows, current airfreight logistics surcharges, and localized retail floors—allowing them to perfectly time farm harvests for optimal market entry.